

WASP-11b

R-0919

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_230819 · created 2026-07-29T08:28:49+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460175.9004 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.051
Transit depth (Rp/Rs)^2	3.01 +/- 1.77 [%]
Orbital Inclination (inc)	90.0 +/- 1.71
Airmass coefficient 1 (a1)	0.42 +/- 0.27
Airmass coefficient 2 (a2)	0.44 +/- 0.33
Scatter in the residuals of the lightcurve fit is	54.53 %
Best Comparison Star	#1 - [418, 142]
Optimal Aperture	0.87
Optimal Annulus	8.72
Transit Duration (day)	0.105 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.174 $\pm$ 0.051 vs 0.131 (deviation 0.84 σ)
Duration fit vs published	0.105 d vs 0.1075 d (deviation 0.21 σ)
Mid-time O–C	8.4 min
Depth SNR	3.4
Residual scatter	54.53 %

Light curve

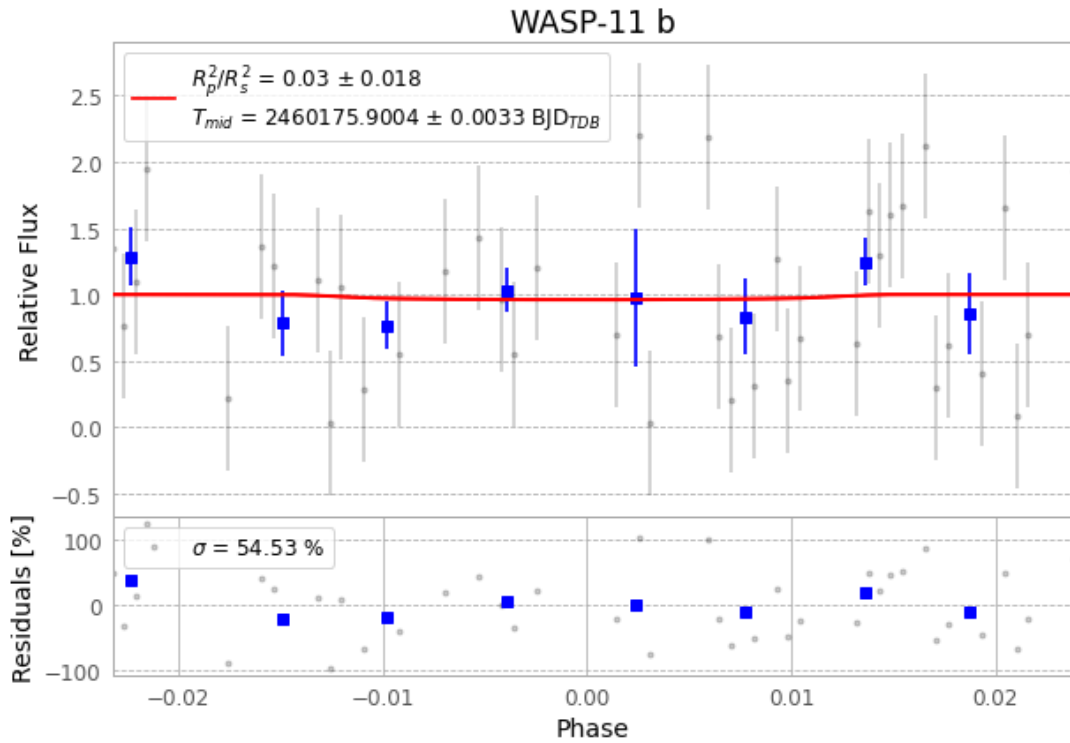


Figure 1 — FinalLightCurve_WASP-11 b_2023-08-19.png (SHA-256 fa4dc1467ac793bd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [454, 420], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d6954bd71d857c47...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

92 FITS frames (60 MB), HATP-10230819071215.FITS → HATP-10230819114515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-230819.log (f2faea516781...), FinalLightCurve_WASP-11 b_2023-08-19.png (fa4dc1467ac7...), FinalLightCurve_WASP-11 b_2023-08-19.csv (e13901bb25b9...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 08:28Z.